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## Problem Set 3: Solutions

### Econ 502: Advanced Microeconomics

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#### Part I: Benchmarks

Throughout:  $P = 150 - Q$ ,  $c = 30$ , so  $a - c = 120$ .

#### Part (a): Perfect competition

Under perfect competition,  $P = MC$ :

$$150 - Q = 30 \implies Q_c = 120, \quad P_c = \$30$$

Consumer surplus is the triangle between the demand curve and the price (see figure below):

$$CS_c = \frac{1}{2}(150 - 30)(120) = \$7,200$$

Producer surplus is zero ( $P = MC$  with constant marginal cost).

|                                                                           |
|---------------------------------------------------------------------------|
| $P_c = \$30, \quad Q_c = 120, \quad CS_c = \$7,200, \quad TS_c = \$7,200$ |
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#### Part (b): Monopoly

Total revenue:  $TR = (150 - Q)Q = 150Q - Q^2$ . Marginal revenue:

$$MR = 150 - 2Q$$

Setting  $MR = MC$ :

$$150 - 2Q = 30 \implies Q_m = 60, \quad P_m = 90$$

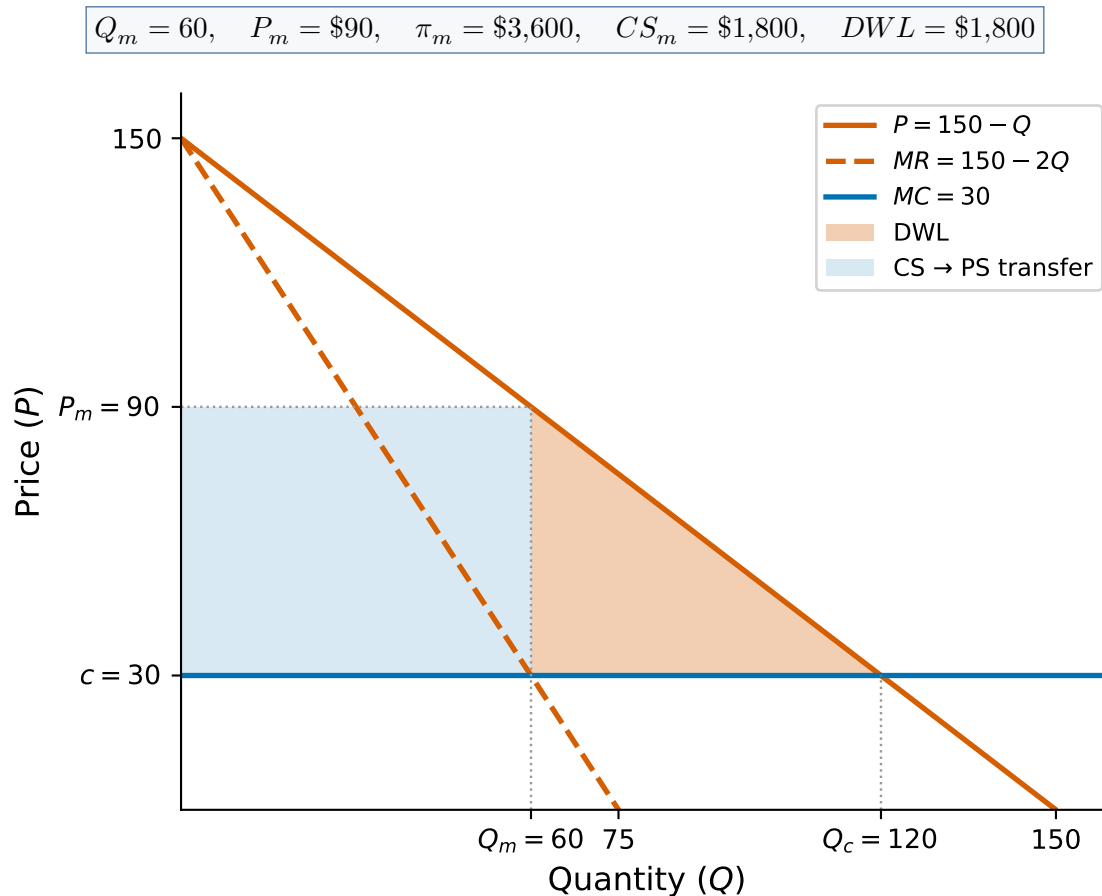
$$\pi_m = (90 - 30)(60) = \$3,600$$

$$CS_m = \frac{1}{2}(150 - 90)(60) = \$1,800$$

$$TS_m = CS_m + \pi_m = 1,800 + 3,600 = \$5,400$$

$$DWL = TS_c - TS_m = 7,200 - 5,400 = \$1,800$$

This equals the triangle:  $\frac{1}{2}(P_m - c)(Q_c - Q_m) = \frac{1}{2}(60)(60) = \$1,800$ .



## Part II: Price Competition

### Part (c): Bertrand

With  $n \geq 2$  firms competing on price, the Nash equilibrium is:

$$p_1 = p_2 = \dots = p_n = c = 30$$

This replicates the **perfectly competitive outcome**:  $P = 30$ ,  $Q = 120$ , zero profits.

**Why?** Suppose all firms charge some  $p > c$ . Any single firm can capture the entire market by cutting its price to  $p - \epsilon$ , roughly multiplying its profit. This undercutting incentive continues until price is driven down to marginal cost. At  $p = c$ , no firm can profitably deviate: raising the price loses all customers, and lowering it means selling below cost.

This is the **Bertrand paradox**: just two firms are enough to eliminate all market power and reproduce the competitive outcome.

### Part III: Quantity Competition

#### Part (d): Cournot duopoly ( $n = 2$ )

Firm 1's profit:

$$\pi_1 = (150 - q_1 - q_2)q_1 - 30q_1 = (120 - q_1 - q_2)q_1$$

First-order condition:

$$\frac{\partial \pi_1}{\partial q_1} = 120 - 2q_1 - q_2 = 0$$

**Best response function:**

$$q_1^*(q_2) = 60 - \frac{q_2}{2}$$

By symmetry,  $q_2^*(q_1) = 60 - q_1/2$ . In the symmetric equilibrium  $q_1 = q_2 = q^*$ :

$$q^* = 60 - \frac{q^*}{2} \implies q^* = 40$$

$$Q^* = 80, \quad P^* = 70, \quad \pi^* = (70 - 30)(40) = \$1,600$$

$$q^* = 40, \quad P^* = \$70, \quad \pi^* = \$1,600 \text{ per firm}$$

#### Part (e): Cournot with $n$ firms

Firm  $i$  maximizes:

$$\pi_i = \left( 150 - q_i - \sum_{j \neq i} q_j \right) q_i - 30q_i$$

First-order condition:

$$120 - 2q_i - \sum_{j \neq i} q_j = 0$$

In a symmetric equilibrium,  $q_j = q^*$  for all  $j$ , so  $\sum_{j \neq i} q_j = (n-1)q^*$ :

$$120 - 2q^* - (n-1)q^* = 0 \implies 120 = (n+1)q^*$$

$$q^* = \frac{120}{n+1}$$

**Part (f): Convergence**

Remember  $c = 30$  and:

$$q^* = \frac{120}{n+1}, \quad P^* = 150 - nq^*, \quad \pi^* = (P^* - c)q^*, \quad \mu = \frac{P^*}{c}$$

| $n$ | Per-firm output<br>( $q^*$ ) | Price ( $P^*$ ) | Per-firm profit<br>( $\pi^*$ ) | Markup ( $\mu$ ) |
|-----|------------------------------|-----------------|--------------------------------|------------------|
| 1   | 60                           | \$90            | \$3,600                        | 3.00             |
| 2   | 40                           | \$70            | \$1,600                        | 2.33             |
| 5   | 20                           | \$50            | \$400                          | 1.67             |
| 10  | 10.9                         | \$40.9          | \$119                          | 1.36             |
| 50  | 2.35                         | \$32.9          | \$6.8                          | 1.09             |

As  $n$  gets large, price gets closer to marginal cost ( $P^* \rightarrow 30$ ) and the markup  $\mu$  approaches 1, replicating the competitive outcome.

**Part IV: Welfare****Part (g): Deadweight loss**

DWL is the triangle between the demand curve and the price, from  $Q^*$  to  $Q_c$ :

$$DWL = \frac{1}{2}(P^* - c)(Q_c - Q^*)$$

Note that

$$Q^* = nq^* = \frac{120n}{n+1}$$

Since  $P^* = 150 - Q^*$ , we can write:

$$P^* = 150 - \frac{120n}{n+1} = \frac{150 + 30n}{n+1}$$

Therefore:

$$DWL = \frac{1}{2} \left( \frac{150 + 30n}{n+1} - 30 \right) \left( 120 - \frac{120n}{n+1} \right) = \frac{7,200}{(n+1)^2}$$

**Comparison:**

- Monopoly ( $n = 1$ ):  $DWL = 7,200/4 = \$1,800$
- Cournot duopoly ( $n = 2$ ):  $DWL = 7,200/9 = \$800$

Going from monopoly to duopoly cuts the DWL by more than half. The DWL shrinks as  $1/(n+1)^2$ , so adding more firms reduces welfare loss rapidly. By  $n = 5$ , DWL is only \$200 (compared to \$1,800 under monopoly).